LESSON 1.4

DIVIDING FRACTIONS

LESSON INTRODUCTION

MA.5.AR.1.3 Solve real-world problems involving division of a unit fraction by a whole number and a whole number by a unit fraction.

LEARNING TARGETS

- I can represent division of fractions with an appropriate model.

BENCHMARK COHERENCE

WORKING TOWARD

MA.6.NSO.2.3 Solve multistep real-world problems involving any of the four operations with positive multidigit decimals or positive fractions, including mixed numbers.

ESSENTIAL QUESTIONS

- How can you create an appropriate model to represent dividing fractions?
- How do you know what the dividend, divisor, and quotient are in a real-world fraction problem?
- How is dividing whole numbers similar or different to dividing fractions?
- What algorithm can we use to divide fractions?

LESSON NARRATIVE

Students explore division of fractions by noting patterns in problems with fractional quotients, divisors, and dividends and creating tape diagrams to represent and understand algorithms. Students collaborate on making sense of tape diagrams, then are guided through real-world problems involving tape diagrams. Students use their understanding to then create their own algorithms for dividing fractions and share their thinking with others. Students show their understanding by flexibly using models and the algorithm to find quotients of fraction division.

COMPONENT SUMMARY

1.4.1 WARM-UP *(~5 MIN.)*

Notice and Wonder with fraction division patterns

1.4.2 COLLABORATION *(10 MIN.)*

Students collaborate to create tape diagrams and equations with fraction division

1.4.3 GUIDED INSTRUCTION *(15 MIN.)*

Guided instruction on modelling the divisor, dividend, and quotient in context and writing corresponding expressions in real-world contexts

1.4.4 COLLABORATION *(10 MIN.)*

Students create, test, and share an algorithm for fraction division

1.4.5 WRAP-UP *(~5 MIN.)*

Use a model to solve fraction division and self-assessment of preferred method

RESOURCES

GLOSSARY

Key Terms:

- N/A

Additional Terms:

- Dividend, Divisor
- Quotient
- Algorithm
- Tape diagram

MATERIALS

- (Optional) Poster paper
- (Optional) Markers or colored pencils
- (Optional) Graph paper

ADDITIONAL PREPARATION

- If using a Gallery Walk setup or poster presentation for 1.4.4 Collaboration, gather poster paper and markers before the lesson so students have access to needed materials.
- Consider having graph paper and colored pencils available for students to use to create tape diagrams throughout the lesson.

TEACHER TIPS

COMMON MISCONCEPTIONS

- Students may not remember which parts of a division problem are the divisor, dividend, or quotient.
- Students may know the rule to divide fractions as keep-change-flip but not understand what that means or why it works.
- Students may struggle with creating an accurate fraction model.
- Students may have difficulty knowing which number is the divisor and which is the dividend in real-world word problems.

THINGS TO CONSIDER

- In this lesson, students will need to create and use representations of fraction division effectively. Be sure to support students in thinking through why and how their fraction tape diagrams represent the expression or context.
- As this unit reviews concepts through scaffolding, it is best not to allow the arithmetic to be the barrier. Consider being flexible with the use of a calculator, as there are more opportunities for fluency with arithmetic in the 6th grade lessons.
- Consider highlighting the mathematicians Charles Babbage and Ada Lovelace and their work with the first machine to make computations.

LITERACY AND LANGUAGE ROUTINES

Notice and Wonder

In 1.4.1 Warm-Up, students are introduced to division patterns with fractional values. Encourage students to share what they notice and wonder about these patterns and how the relationship between the dividend, divisor, and quotient changes based on the values of each in the pattern. Consider writing these ideas in a shared space for students to revisit during 1.4.4 Collaboration.

MATHEMATICAL THINKING AND REASONING (MTR) STANDARD

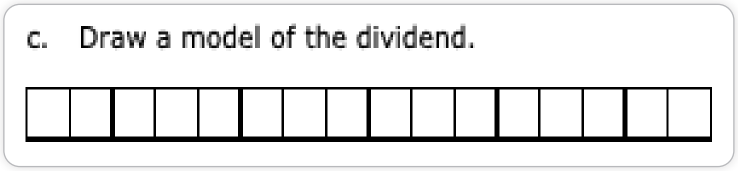
MA.K12.MTR.5.1 Patterns and Structure

Lesson 4 positions students as experts of their own observations. Students do not memorize and apply steps to divide fractions. Rather, students must make observations, notice patterns, hypothesize standard algorithms based on those patterns, and apply to verify in the context of division of fractions.

DIFFERENTIATE INSTRUCTION

In 1.4.3 Guided Instruction invites students to draw and label their model to show what is happening when they divide fractions so they can ask themselves if their solution makes sense. Once students find the values of the expression, ask students if their solution is reasonable for the context. This will help them refine their understanding of the concept of fractional division.

In 1.4.3 Guided Instruction, consider using different colors and graph paper to model how to create tape diagrams and explain the steps/thinking through a Think Aloud. This auditory and visual support may serve as an entry point and guide students to reason about fraction division and make sense of models.



1.4.1 WARM-UP: NOTICE AND WONDER

Component Overview: *Give students three to five minutes to individually write down what they notice and wonder about the patterns of division for questions 1 and 2, then come together as a class to engage in Notice and Wonder literacy routines to share ideas and record them in a shared space.*

TEACHER MOVES

Notice and Wonder

Use Notice and Wonder literacy routines to engage in a class discussion on the patterns with the quotients. As students share, consider writing the students' ideas as a list on a shared classroom space, like a whiteboard or chart paper, for students to see all responses in one place and make connections.

ENRICHMENT

Position students to recognize how the quotients are affected by whether the dividend is a fraction or a whole number and whether the divisor is getting smaller or larger. Ask questions to support this reasoning, like:

- What do you notice about the values of the dividends, divisors, and the quotients in the different boxes?
- What do you notice about the quotients when the divisors get larger, like in the yellow and pink boxes? What happens to the quotient pattern?
- What do you notice about the quotients when the divisors get smaller, like in the blue and green boxes? What happens to the quotient pattern?

1.4.1 Warm-Up: Notice and Wonder

There are four patterns of division. The last problem in the pattern does not have a quotient.

$50 \div 100 = \frac{1}{2}$ $50 \div 10 = 5$ $50 \div 1 = 50$ $50 \div \frac{1}{10} = 500$	$\frac{1}{2} \div 1 = \frac{1}{2}$ $\frac{1}{2} \div 10 = \frac{1}{20}$ $\frac{1}{2} \div 100 = \frac{1}{200}$
$\frac{1}{100} \div 1 = \frac{1}{100}$ $\frac{1}{100} \div 10 = \frac{1}{1000}$ $\frac{1}{100} \div 100 = \frac{1}{10{,}000}$	$200 \div 100 = 2$ $200 \div 10 = 20$ $200 \div 1 = 200$ $200 \div \frac{1}{10} = 2{,}000$

1. **What do you notice about the patterns?**

   Answers will vary. I notice that each new quotient has an additional zero digit. I notice that dividing by a fraction less than one results in a larger quotient.

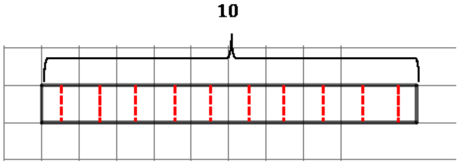
2. **What do you wonder about the patterns?**

   Answers will vary. I wonder how to divide by other fractions. I wonder if this pattern holds when dividing by a number other than 10.

3. **Use the pattern to guess what the missing quotient is for each pattern. Write your guess in the box.**
   Answers shown in boxes.

1.4.2 Collaboration: Visual Models of Division

1. A tape diagram is shown.



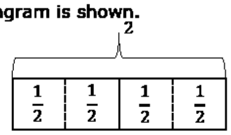
   a. Use the diagram to show how many groups of $\frac{1}{2}$ are in 10.
   20 *groups. Answer shown on diagram.*

   b. Write the division problem represented by the tape diagram.

dividend		divisor		quotient
10	÷	$\frac{1}{2}$	=	20

   c. Discuss with your partner how the tape diagram shows that dividing a whole number by a fraction less than 1 will result in a quotient that is greater than the dividend.

2. A tape diagram is shown.



   Write the division equation that is represented by the tape diagram.

dividend		divisor		quotient
2	÷	$\frac{1}{2}$	=	4

1.4.2 COLLABORATION: VISUAL MODELS OF DIVISION

Component Overview: *In this Collaboration, students explore how to represent fraction division using tape diagram models. Students work through the activity together at their own pace while monitoring their progress and using the scaffolded questions provided as opportunities for deeper discussion.*

TEACHER MOVES

For students who are having difficulty with division with fractional values, try rephrasing the division statement using a question. For example, in question 1b, ask students:

- How many groups of ½ are in 10?

For question 2, ask students:

- How many groups of ½ are in 2?

After asking the question, prompt students to point out how they can see these groups within their tape diagrams and physically touch and count those groups to confirm their solutions.

1.4.2 COLLABORATION: VISUAL MODELS OF DIVISION (CONTINUED)

ENRICHMENT

Activate prior knowledge of division with whole numbers and division with fractional parts.

- How do you know which is the dividend, divisor, and quotient?
- If needed, provide whole-number examples to help students recognize which value is being divided (dividend) by which value (divisor), then reattempt with the fractional value in question 1b.

TEACHER MOVES

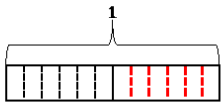
Notice in question 3 that the dividend is now a fractional value. As collaborative groups begin question 3, help students recognize how this changes the division problem by asking:

- What do you notice about the dividend and divisor in question 2 versus question 3? What is the same? What is different?
- How does dividing by a fraction differ from dividing by a whole number?
- What do you think will happen to the quotient if the dividend is a fraction? What does that mean about the value we are dividing?
- How does having a fractional dividend change how you create your tape diagram?

QUESTIONING

For early finishers, have students pick any of the four equations from this section and write a different division equation using equivalent fractions that has the same quotient.

3. **The tape diagram shows 6 groups in $\frac{1}{2}$.**

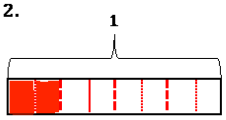


   a. **Draw the 6 groups for the other $\frac{1}{2}$.**
   Answer shown on diagram.

   b. **Complete the explanation by writing a value for each blank.**

   The tape diagram shows that $\frac{1}{2} \div 6 = \frac{1}{12}$ because 1 whole is first divided into two parts or $\frac{1}{2}$. Then, each half is divided into 6 groups. Since the tape diagram represents the value of 1 and it has now been divided into 12 parts, each part is represented by the fraction, $\frac{1}{12}$.

4. **Work with your partner to use the tape diagram to show $\frac{1}{4} \div 2$.**



   The tape diagram shows that the quotient of $\frac{1}{4} \div 2$ is $\frac{1}{8}$.
   Answer shown on diagram.

1.4.3 Guided Instruction: Dividing Fractions

1. The Blue Palmetto Café at Bok Tower Gardens has several items available for purchase so guests can enjoy a meal after touring the gardens. Before the gardens opened for the day, the chef took inventory of the food on hand.

Food	Number Available for Purchase
Turkey Cranberry Wrap	12
Vegetarian Wrap	$\frac{1}{2}$ of a wrap
Peanut Butter and Jelly Sandwich	5
Iced Tea	2 gallons
Vanilla Ice Cream	$\frac{1}{4}$ of a gallon

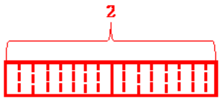
a. Each cup of iced tea is approximately $\frac{1}{8}$ of a gallon. Write the division expression that represents how many cups of iced tea the chef had on hand that day.

$2 \div \frac{1}{8}$

b. The dividend is
- o $\frac{1}{8}$ of a gallon.
- o 2 gallons.

When drawing a model to represent the equation, first draw the ***dividend***, or the amount being divided.

c. Draw a model of the dividend.



1.4.3 GUIDED INSTRUCTION: DIVIDING FRACTIONS

Component Overview: *Students identify the dividend, divisor, and quotient within the context of a restaurant inventory, using those values to identify the number of groups and value of each group in real-world division problems. Work together as a class as students are guided through this instruction, continuing to ask students to justify their understanding using the context of the problem.*

DIFFERENTIATED INSTRUCTION

Consider providing graph paper for added visual support when creating tape diagrams in this section. Use a Think Aloud approach to model and explain thinking as you show students how to model the dividend and divide by the divisor using a tape diagram. Strengthen math vocabulary by labelling the divisor, dividend, and quotient in the modelled tape diagram.

ENRICHMENT

For question 1c, encourage students to make sense of the tape diagram and fractional division by asking:

- What fraction does each box represent? How can you tell? How does this relate to the real-world context of iced tea?

1.4.3 GUIDED INSTRUCTION: DIVIDING FRACTIONS (CONTINUED)

TEACHER MOVES

To strengthen the connection between division vocabulary and the fraction-division context, have students label their expressions with the glossary words as they move through this section. Refer to Unit Guide for more information.

ENRICHMENT

For question 2c, ask:

- How are the number of students represented on your diagram?
- How can you tell that your solution is reasonable given the context?

Then, partition the dividend into equal groups determined by the *divisor*, or the number that tells the size of each group.

d. In this problem, the divisor is
- ○ $\frac{1}{8}$ of a gallon.
- ○ 2 gallons.

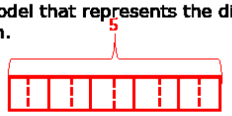
e. Complete your model above to show the dividend partitioned into equal groups that are the size of the divisor.
Answer shown on diagram.

f. Use your model to determine the number of cups of iced tea that can be served before the café runs out.
16 cups

2. A local pre-school decided to go on a field trip to the gardens. The chef knew that for the kids' meals, each child would receive $\frac{1}{2}$ of a peanut butter and jelly sandwich.

a. Write the division expression that represents how many kids' lunches could be made with the number of peanut butter and jelly sandwiches on hand that day.
$5 \div \frac{1}{2}$

b. Draw a model that represents the division expression.

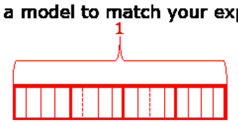


c. Use the model to determine the number of students the chef can serve.
10 lunches

3. Veer's family was very hot after an afternoon in the gardens. They decided to enjoy some ice cream before they left to go home. All 4 family members ordered vanilla ice cream, causing the chef to run out. How much ice cream did each person get if they each got an equal amount?

   a. Write the division expression that represents the context.
   $\frac{1}{4} \div 4$

   b. Draw a model to match your expression.



   c. How much ice cream did each person get?
   $\frac{1}{16}$ *of a gallon*

1.4.3 GUIDED INSTRUCTION: DIVIDING FRACTIONS (CONTINUED)

DIFFERENTIATED INSTRUCTION

Consider providing different colored pencils or highlighters as a visual support for seeing the divisor, dividend, and quotient as students draw a model to match the expression in question 3.

For example, use one color to draw a box around every four parts of the tape diagram, each representing one whole, so students can see the shaded portion represents ¼ of a whole (¼ is the dividend). Then use a different color to count and label the total boxes in the diagram (16) so students can see that one out of the 16 boxes are shaded and that the shaded portion of the whole bar diagram represents the quotient.

QUESTIONING

The focus of this section is to use tape diagrams to represent fraction division, though there are other methods to model fraction division.

Consider prompting early finishers or students with a strong grasp on fraction division to represent the division equation in question 1 with a different diagram. Encourage creativity and celebrate and recognize unique responses.

1.4.4 COLLABORATION: AN ALGORITHM FOR DIVIDING FRACTIONS

Component Overview: *Pair students as they revisit a similar quotient pattern from the Warm-Up. With the Guided Instruction and discussions from the lesson, students now recognize the fraction-division examples and create an algorithm of dividing fractions that they will test and share with others.*

DIFFERENTIATED INSTRUCTION

Consider providing students with poster paper as they develop their algorithms in questions 1a-d, and use a Gallery Walk routine or have groups share their algorithm using the poster paper as part of their presentation in questions 1e-1f.

TEACHER MOVES

As students study the patterns, prompt them to draw arrows to the two sets of boxes showing relationships or connections between the patterns. Ask:

- Where do you see similarities between the parts of the fraction terms across the dividend, divisor, and quotient?

1.4.4 Collaboration: An Algorithm for Dividing Fractions

The boxes show the division equations that have been represented in the lesson so far.

$$10 \div \frac{1}{2} = 20 \qquad 2 \div \frac{1}{2} = 4 \qquad 1 \div \frac{1}{8} = 8 \qquad 5 \div \frac{1}{2} = 10$$

$$\frac{1}{2} \div 6 = \frac{1}{12} \qquad \frac{1}{4} \div 2 = \frac{1}{8} \qquad \frac{1}{4} \div 4 = \frac{1}{16}$$

Keeping in mind that a whole number can be written as a fraction, rewrite the division equations in the two boxes

$$\frac{10}{1} \div \frac{1}{2} = 20 \qquad \frac{2}{1} \div \frac{1}{2} = 4 \qquad \frac{1}{1} \div \frac{1}{8} = 8 \qquad \frac{5}{1} \div \frac{1}{2} = 10$$

$$\frac{1}{2} \div \frac{6}{1} = \frac{1}{12} \qquad \frac{1}{4} \div \frac{2}{1} = \frac{1}{8} \qquad \frac{1}{4} \div \frac{4}{1} = \frac{1}{16}$$

Charles Babbage is credited with inventing the first mechanical computer. Ada Lovelace was the first to recognize that Babbage's machine had applications beyond making calculations. She published the first *algorithm* intended to be carried out by such a machine. As a result, she is sometimes regarded as the first to recognize the full potential of a "computing machine" and the first computer programmer.

1. **Work with your partner on an algorithm for dividing fractions.**
   a. **Write the algorithm as a set of rules to be followed. Use mathematical terms such as *divisor* and *dividend*.**

   Your algorithm should meet the following criteria:
   - **The algorithm uses mathematical terms.**
   - **The algorithm works for our test problems.**
   - **The algorithm is clear and can help students learn the process of dividing fractions.**

   b. **Test your algorithm using the problems shown. If necessary, adjust your algorithm.**

   $8 \div \frac{7}{3} = \frac{24}{7}$

   $4 \div \frac{2}{3} = \frac{12}{2}$

   $5 \div \frac{4}{9} = \frac{45}{4}$

   $\frac{2}{3} \div 6 = \frac{2}{18}$

   $\frac{3}{4} \div 2 = \frac{3}{8}$

   $\frac{5}{7} \div 4 = \frac{5}{28}$

   c. **If your algorithm worked and did not need adjustment, then show the work that proves that it worked. If your algorithm did not work, remember making mistakes helps you learn. Adjust your algorithm and write the new steps.**

   d. **Test your algorithm again or use your new algorithm with the problems shown.**

   $\frac{7}{3} \div \frac{7}{3} = 1$

   $\frac{4}{5} \div \frac{1}{3} = \frac{12}{5}$

   $\frac{1}{2} \div \frac{1}{5} = \frac{5}{2}$

1.4.4 COLLABORATION: AN ALGORITHM FOR DIVIDING FRACTIONS (CONTINUED)

ENRICHMENT

To make the connection between the opening of this activity and this first problem, prompt students to use the relationships outlined in the Warm-Up to create rules that describe what parts of each fraction divisor should operate with parts of the fraction dividend. Ask:

- How do the Warm-Up patterns relate to these patterns?
- How can they help you determine an algorithm for dividing fractions?

TEACHER MOVES

Prompt student discussions by asking how they could use whole numbers to test the algorithm, remembering that a whole number is a fraction with a denominator of one. If students explain without using mathematical terms like “divisor” and “dividend,” rephrase to clarify using correct math vocabulary.

DIFFERENTIATED INSTRUCTION

If students are creating posters to present their algorithms, be sure students have completed through question 1d before starting on their poster. Encourage them to use words, symbols, and numbers to describe their algorithm, prove that it meets the criteria outlined in question 1a, solve worked examples for each of the questions in 1d, and optionally draw tape diagram models as evidence that the algorithm they created works.

1.4.4 COLLABORATION: AN ALGORITHM FOR DIVIDING FRACTIONS (CONTINUED)

TEACHER MOVES

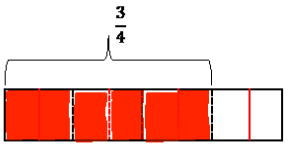
As students exchange the algorithm or optionally present using the poster paper, provide the following questions or sentence stems to help the groups engage in a math discussion about the algorithms:

- "Using your algorithm, could you clarify how to…?"
- "I think your algorithm is accurate because…"
- "One part I would like some more information on is…"
- "How would your algorithm work with ___ divided by ___?"
- "How do you know your algorithm works for dividing all fractions?"

1.4.5 Wrap-Up: Ticket Out the Door

1. Complete the following.
   a. Select what you think is true for the answer to the question: How many groups of $\frac{1}{8}$ are in $\frac{3}{4}$.
      - ☐ The number of groups is larger than $\frac{1}{8}$.
      - ☐ The number of groups is larger than $\frac{1}{8}$ but smaller than $\frac{3}{4}$.
      - ☐ The number of groups is larger than $\frac{3}{4}$.

   b. Using a model can help visualize mathematics. Use the tape diagram to determine the number of groups of $\frac{1}{8}$ that are in $\frac{3}{4}$.



   c. Use an algorithm from class today to find $\frac{3}{4} \div \frac{1}{8}$. Show your work.
      6

2. Check all that apply to you.
   - ☐ I prefer to use models for dividing fractions.
   - ☐ I prefer to use an algorithm for dividing fractions.
   - ☐ I do not mind using a model or an algorithm for dividing fractions.

   Answers will vary.

e. Exchange your algorithm with another pair of students.
- In the first column, write the names of the students who wrote the algorithm.
- In the second column, write your summary of their algorithm.
- In the third column, use their algorithm to test any two of the division equations.

Algorithm Creators	My Summary of Their Algorithm	My Test of Their Algorithm
	Answers will vary.	

f. Check the criteria that this algorithm met.
- ☐ The algorithm uses mathematical terms.
- ☐ The algorithm works for my test problems.
- ☐ The algorithm is clear and could help students know the process of dividing fractions.

1.4.5 WRAP-UP: TICKET OUT THE DOOR

Component Overview: *Students individually show learning by using a model to solve a fraction-division problem, then reflect on which approach for dividing fractions they prefer. Use all or parts of this exercise as a formative assessment to gather data on student progress from this lesson. Students should **not** use a calculator to complete the Wrap-Up.*

DIFFERENTIATED INSTRUCTION

Consider having students draw a tape diagram to model question 1a to support reasoning and as further evidence of student learning.